Note Title

Unit 1.2 The absolute Value. (p. A6 – A8)

2/5/2018

$|a|$ is the distance between a and zero on the number line.

$|a| \geq 0$ for all $a \in \mathbb{R}$.

Definition:

$|x| =$

Also: $|f(x)| =$

$|-3| =$

$|-a| =$

$|-x^2| =$

Properties of the absolute value: (p. A7)

1) $|ab|$

2) $\left|\frac{a}{b}\right|$

3) $|a^n|$

$\sqrt{a^2}$

Graph of $y = |x|$. $y = |x| + 2$.

$y = |x^2 - 1|$.

Solving equations and inequalities with absolute values

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NB: Here we use the theorem on p. A7.

Theorem: Suppose $a>0$. Then:

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Also: Suppose $a>0$. Then:

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Examples:

1) Solve for x if $|x+9|=4$

2) $|x^2-9| = 1$

3) $|x+1| = -1$

4) $|x-4| \leq 4$

5) $|2x-3| \geq 6$

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6) $|x-1| < -4$ $\qquad$ $|x-1| > -4$

7) $\dfrac{x}{|x-2|} > 0$

Compare with solution of: $\left|\dfrac{x}{x-2}\right| > 0$

8) $\dfrac{x^2-x}{|x|} \geq 0$

Compare with solution of $\left|\dfrac{x^2-x}{x}\right| \geq 0$

9) $|x+3| = |2x+1|$ $\qquad$ NB: Test your answer!!

10) $|2x-4| \geq 8$

11) $\dfrac{|x-6|}{x^2-x} \leq 0$

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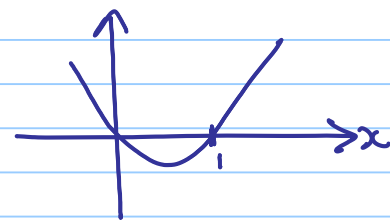
$$\frac{|x-6|}{x^2-x} \le 0$$

Consider : $\frac{|x-6|}{x^2-x} = 0$ or $\frac{|x-6|}{x^2-x} < 0$

$\Longrightarrow$ $x=6$ or $x^2-x<0$



$x \in (0,1)$

$\Longrightarrow x \in (0,1) \cup \{6\}$.

Solution of : $\left|\frac{x-6}{x^2-x}\right| \le 0$?